MiniMalloc: A Lightweight Memory Allocator for Hardware-Accelerated Machine Learning

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The rapid growth in ML **model architectures** has been enabled by a corresponding evolution in **machine architectures**.

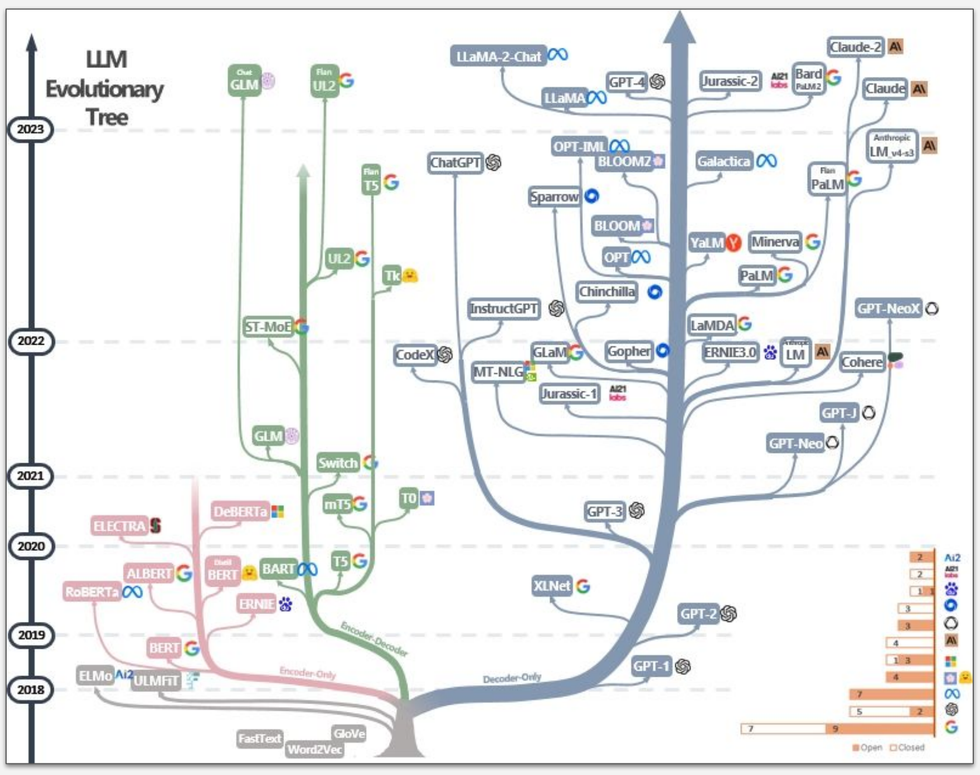
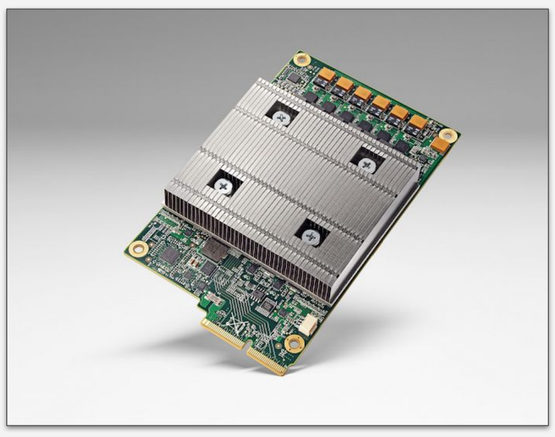
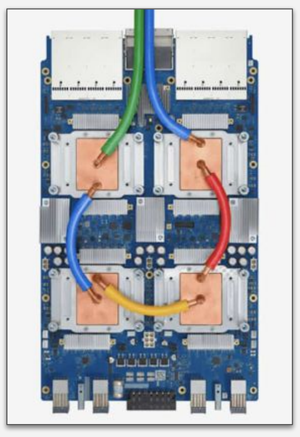


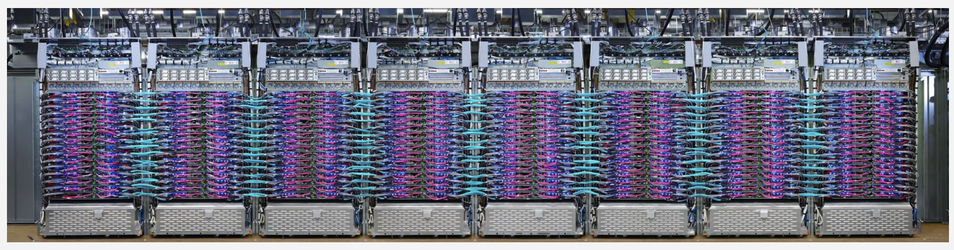
Image Credit: "A Survey on ChatGPT and Beyond", Yang et al., 2024



Tensor Processing Unit board



Cloud TPU device



A Cloud TPU v3 Pod

Image Credit: cloud.google.com/blog/products/ai-machine-learning/

The algorithmic challenges faced by a production-class compiler are significant, and include (among other NP-hard problems) ***static memory allocation***.

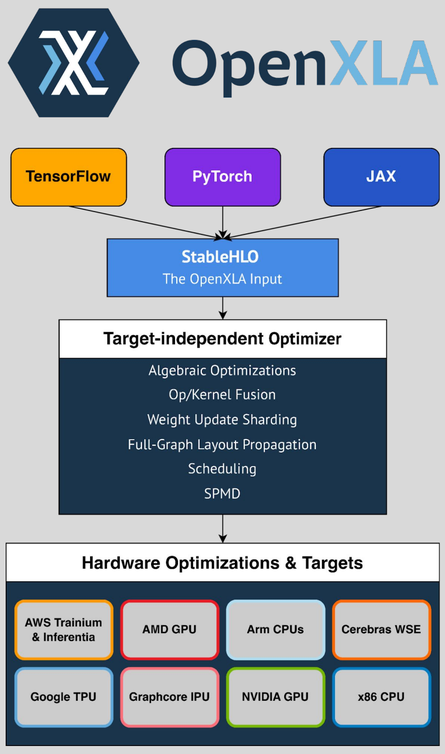


Image Credit: opensource.googleblog.com

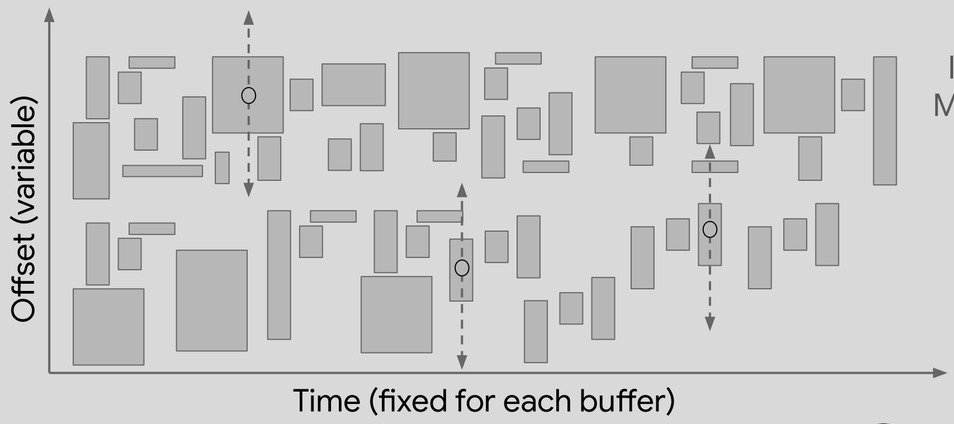
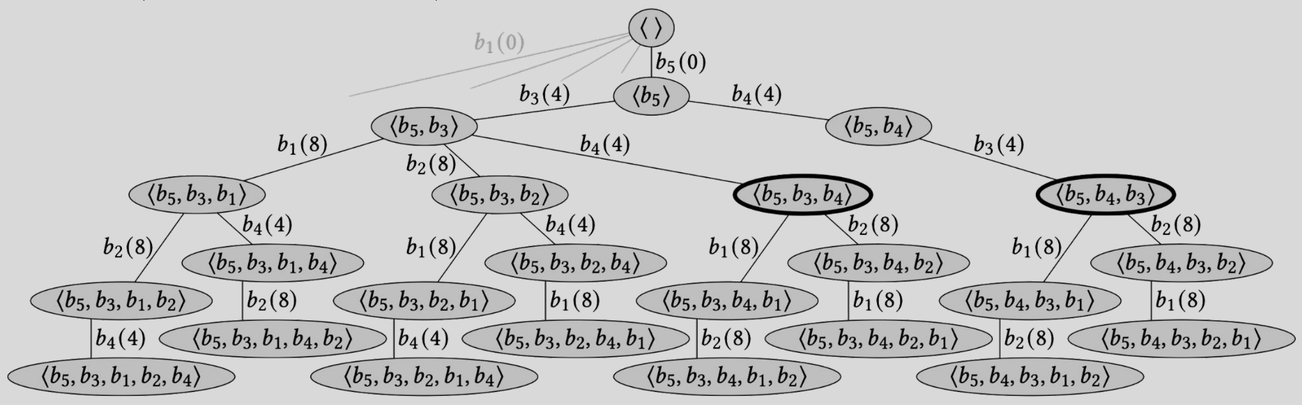
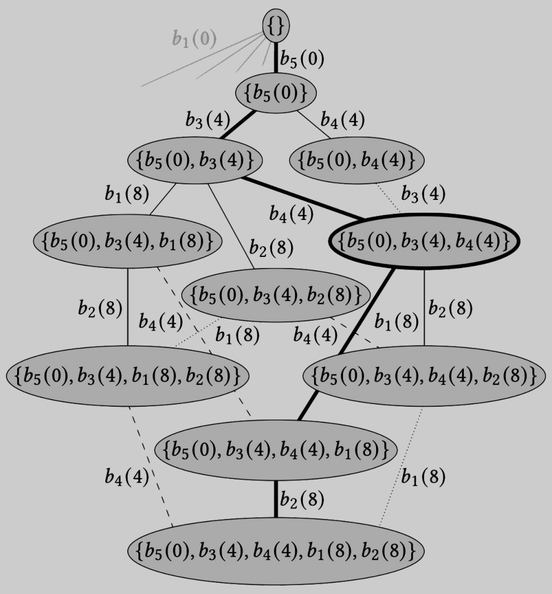


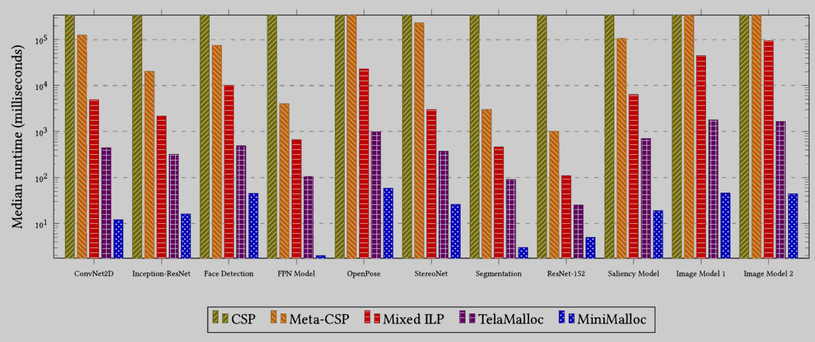
Image Credit: "TelaMalloc: Efficient On-Chip Memory Allocation", Maas et al., ASPLOS 2023

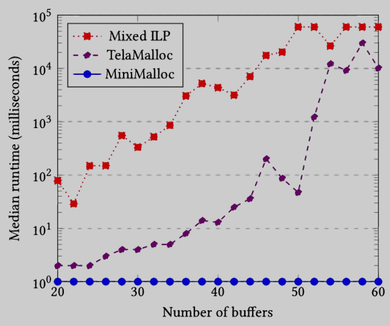


MiniMalloc combines several techniques to reduce the search space, leading to *orders-of-magnitude* performance improvement on many benchmarks.

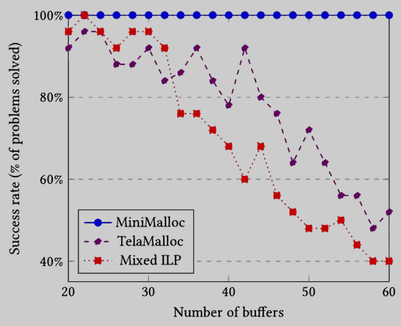


The Search Lattice

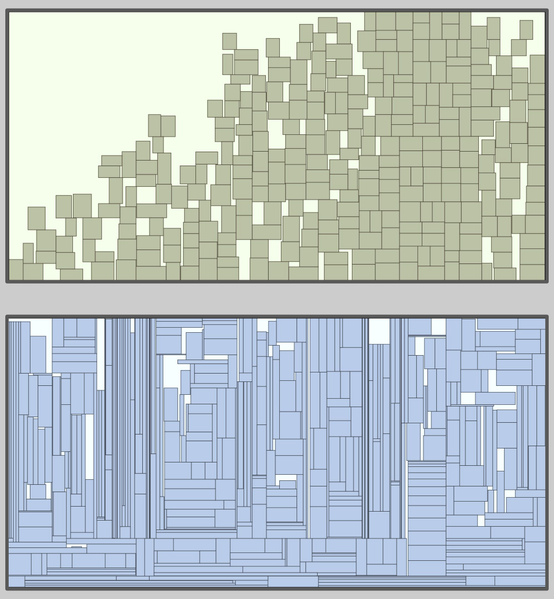




(a) Median runtime of the memory allocators



(b) Success rate of the memory allocators



Examples of MiniMalloc's Allocations